

# ECON 108

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INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

# Real gdp and the price level in the long run

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## CHAPTER VII

# CHAPTER OUTLINE

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Output Growth and the Long-Run Aggregate Supply Curve

Total Expenditure and Aggregate Demand

Shift in the Aggregate Demand Curve

Long Run Equilibrium and the Price Level

Causes of Inflation

# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

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## Aggregate supply

- The aggregate supply is the total of all planned production for the economy

# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

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## Long-run aggregate supply curve

- Is a vertical line representing the real value of final goods and services produced in the economy after full adjustment has occurred
- Input prices fully adjust to changes in output prices
- Suppliers have no incentive to increase output
- Unemployment is at the natural rate
- Determined by endowments and technology (or existing resources)

# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

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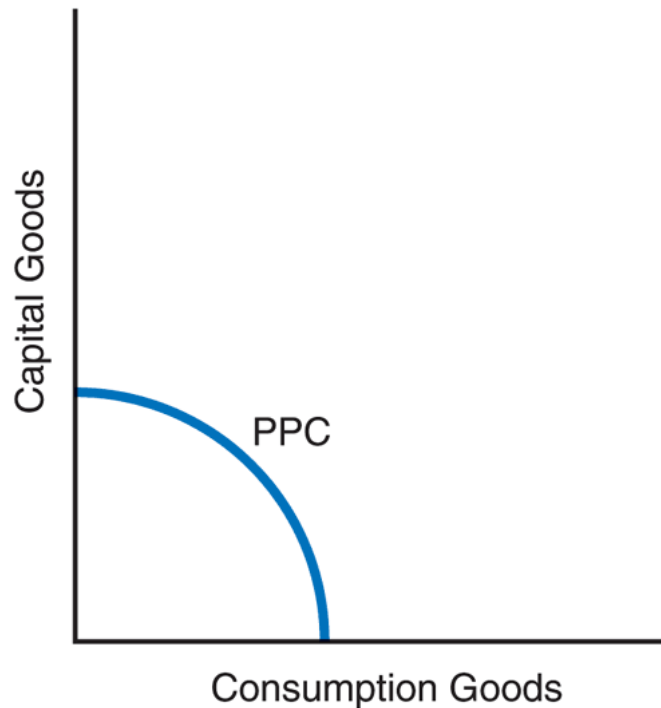
## Long-run aggregate supply curve

- Measured in Base year dollars
- **Endowments** are the various resources in an economy, including both physical resources and resources such as ingenuity and creativity as well as management skills

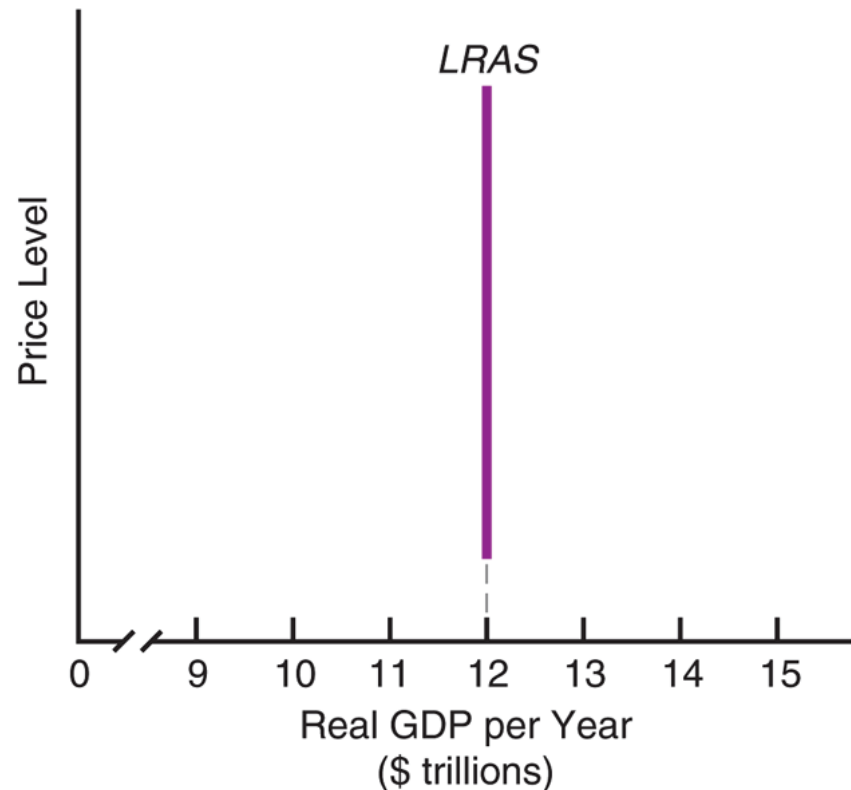
# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

## Long-run aggregate supply curve

Panel (a)



Panel (b)



# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

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## Long-run aggregate supply curve

- Can also be seen as the real GDP of the economy under conditions of full employment
- Also called *full employment real GDP*

# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

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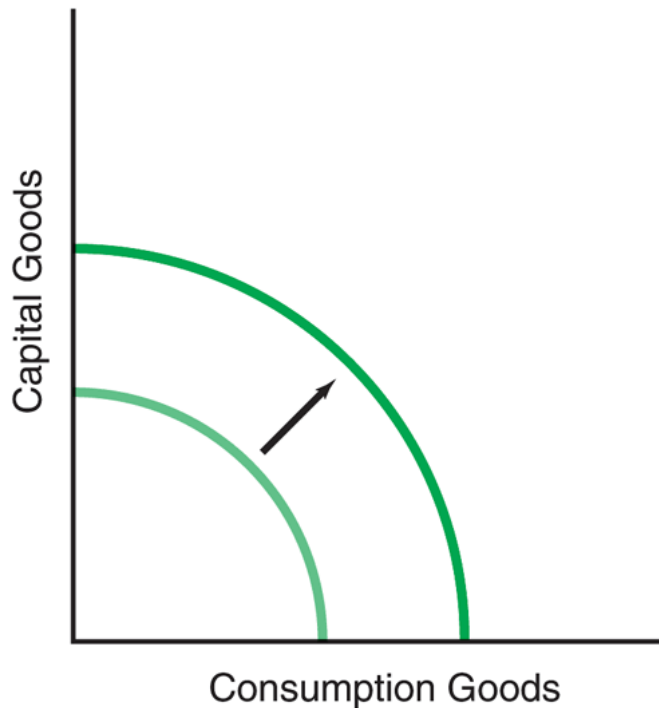
## Economic Growth

- Growth is shown by outward shifts of either the production possibilities curve or the LRAS curve caused by:
  - Growth of population and the labor-force participation rate
  - Capital accumulation
  - Improvements in technology

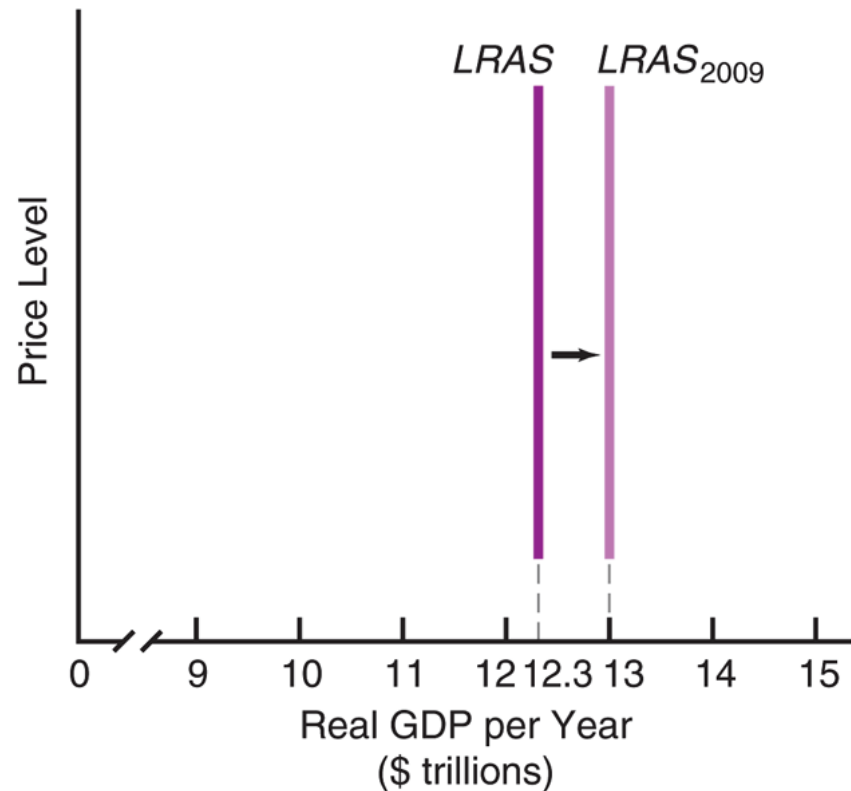
# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

## Economic Growth

Panel (a)

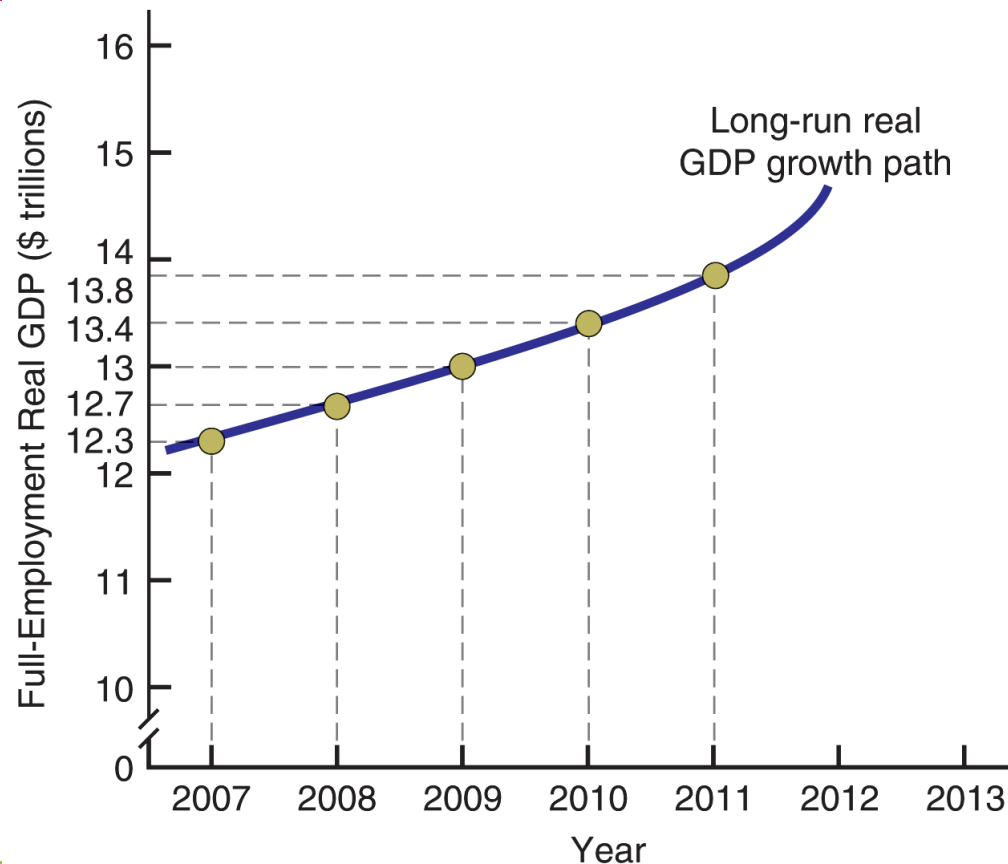


Panel (b)



# OUTPUT GROWTH AND THE LONG-RUN AGGREGATE SUPPLY CURVE

## Economic Growth



# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Aggregate Demand

- The aggregate demand is the total of all planned expenditure in the entire economy

# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Questions

- What determines the total amount that individuals, governments, firms, and foreigners want to spend?
- What determines the equilibrium price level?

# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Aggregate Demand Curve

- The aggregate demand curve shows planned purchase rates for all final goods and services in the economy at various price levels, all other things held constant

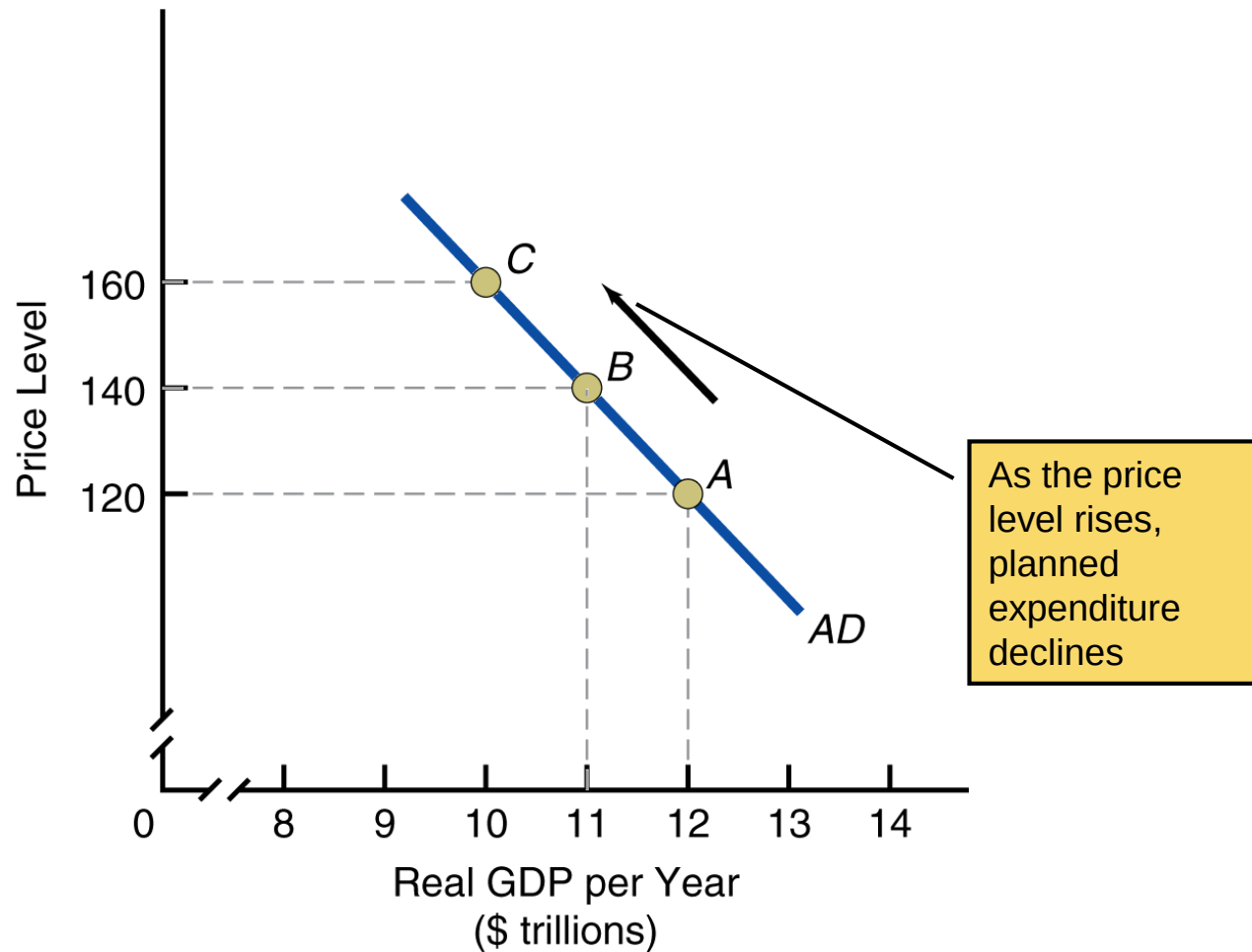
# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Aggregate Demand Curve versus Demand Curve

- When the **aggregate demand** curve is derived, we are looking at the entire circular flow of income and product.
- When a **demand** curve is derived, we are looking at a single product in one market only.

# TOTAL EXPENDITURE AND AGGREGATE DEMAND



# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Slope of the demand curve

- As the price changes, the planned real expenditure changes due to three effects:
  - The real-balance effect (AKA wealth effect)
  - The interest-rate effect
  - The open economy effect

# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Real-balance effect

- The change in the real value of money balances when the price level changes
  - An increase in the price level reduces the value of money balances reducing planned expenditure.
  - A decreases in the price level increases the value of money balances increasing planned expenditure.

# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Interest Rate effect

- The change in the interest rate when the price level changes.
  - An increase in the price level increases the interest rate reducing planned expenditure.
  - A decreases in the price level decreases the interest rate increasing planned expenditure.

# TOTAL EXPENDITURE AND AGGREGATE DEMAND

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## Open Economy effect

- The change in the demand of domestic products by foreigners when the price level changes
  - An increase in the price level decreases the demand of foreigners for domestic products reducing planned expenditure.
  - A decreases in the price level increases the demand of foreigners interest rate increasing planned expenditure.

# SHIFTS IN THE AGGREGATE DEMAND CURVE

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- Any non-price-level change that increases aggregate spending (on domestic goods) shifts *AD* to the right.
- Any non-price-level change that decreases aggregate spending (on domestic goods) shifts *AD* to the left.

# SHIFTS IN THE AGGREGATE DEMAND CURVE

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## Changes That Cause an Increase in Aggregate Demand

An increase in the amount of money in circulation  
Increased security about jobs and future income  
Improvements in economic conditions in other countries  
A reduction in real interest rates (nominal interest rates corrected for inflation) not due to price level changes  
Tax decreases  
A drop in the foreign exchange value of the dollar

## Changes That Cause a Decrease in Aggregate Demand

A decrease in the amount of money in circulation  
Decreased security about jobs and future income  
Declines in economic conditions in other countries  
A rise in real interest rates (nominal interest rates corrected for inflation) not due to price level changes  
Tax increases  
A rise in the foreign exchange value of the dollar

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# Exchange Rates

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## Foreign Exchange Rate

In US is the number of dollars needed to buy one unit of foreign currency.

- Example: the exchange rate Dollar Euro in US is  $E=1.25$ .
- An increase in the exchange rate (depreciation of the dollar) means that the exchange value of the dollar declines and vice versa

# Exchange Rates

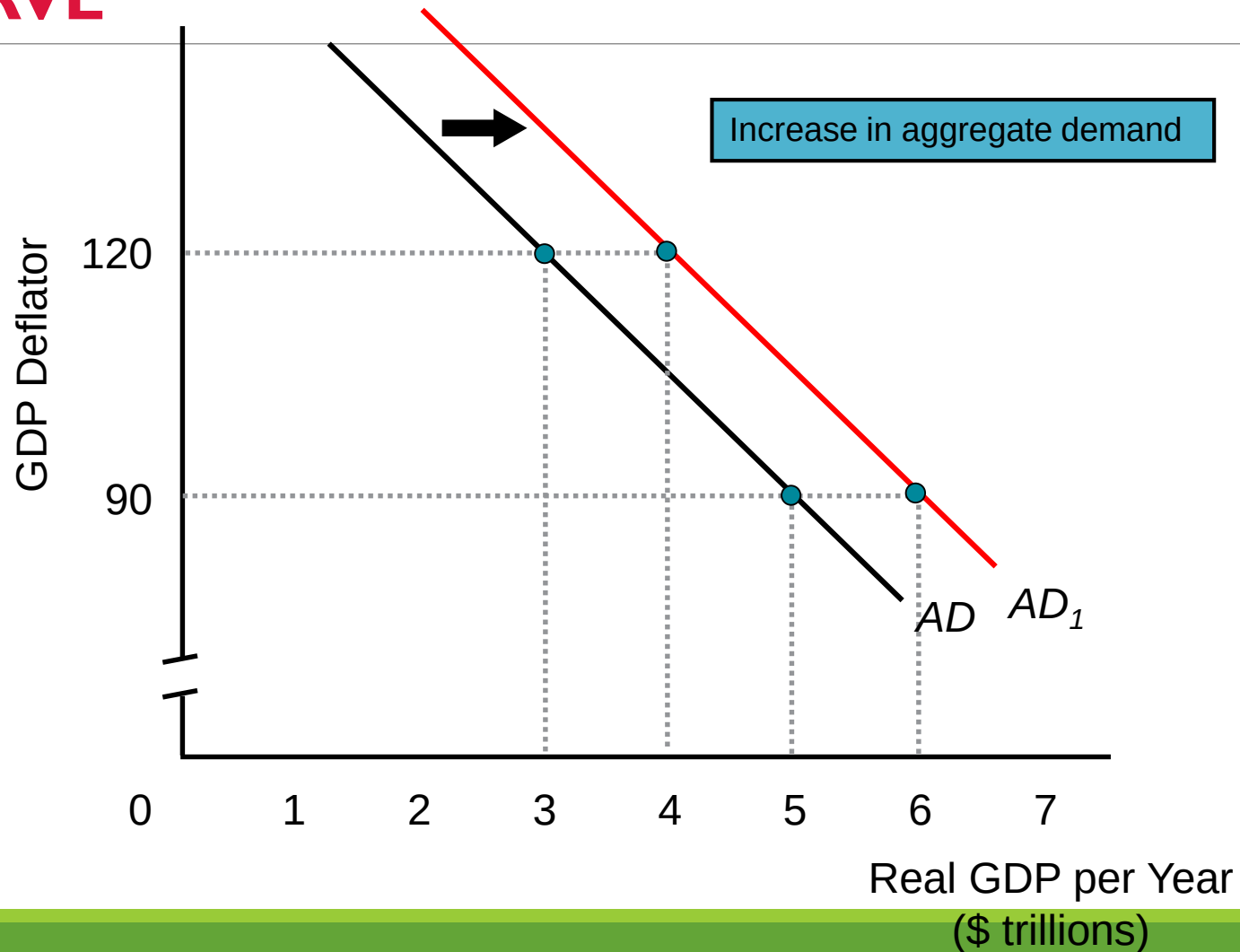
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## Foreign Exchange Rate

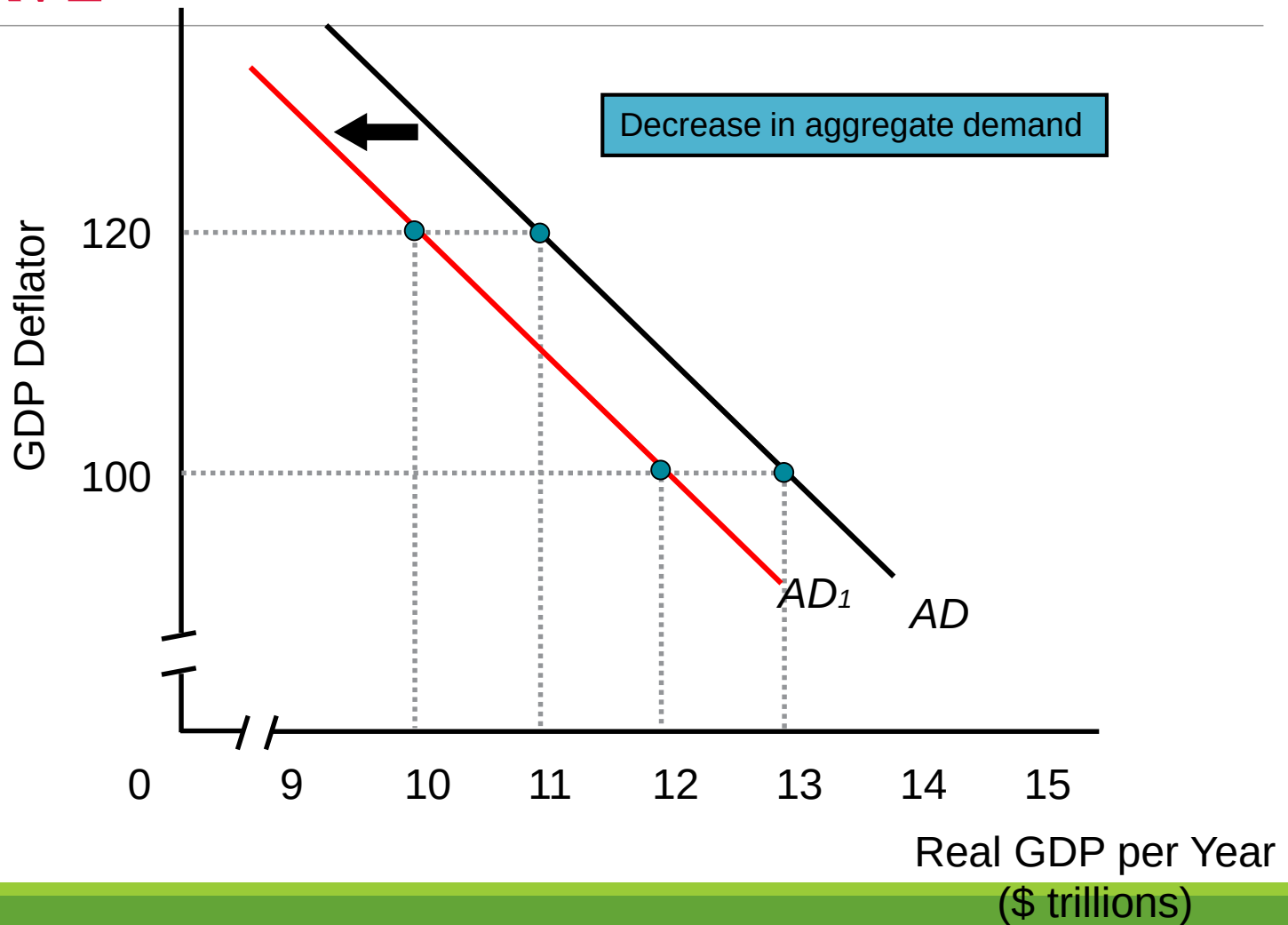
When the exchange rate increases, the exchange value of the dollar depreciates (the dollar becomes weaker).

- Exports tend to increase.
- Imports tend to decline.

# SHIFTS IN THE AGGREGATE DEMAND CURVE



# SHIFTS IN THE AGGREGATE DEMAND CURVE

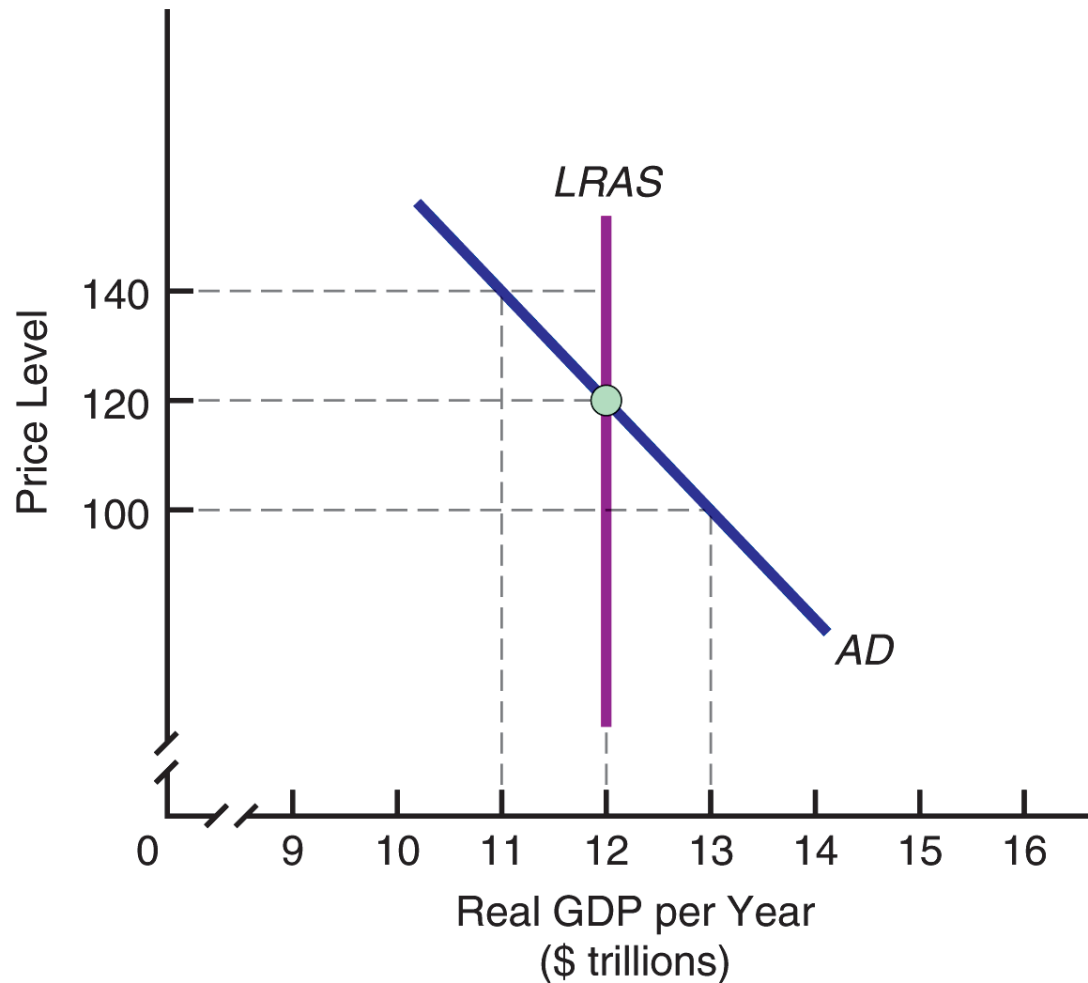


# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL

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- For the economy as a whole, long-run equilibrium occurs at the price level where the aggregate demand curve (*AD*) crosses the long-run aggregate supply curve (*LRAS*).

# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL



# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL

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- The effects of economic growth on the price level
- Economic growth and secular deflation

# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL

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## Secular deflation

- A persistent decline in prices resulting from economic growth in the presence of stable aggregate demand
- An increase in *LRAS* will, *ceteris paribus*, result in a decrease in the price level

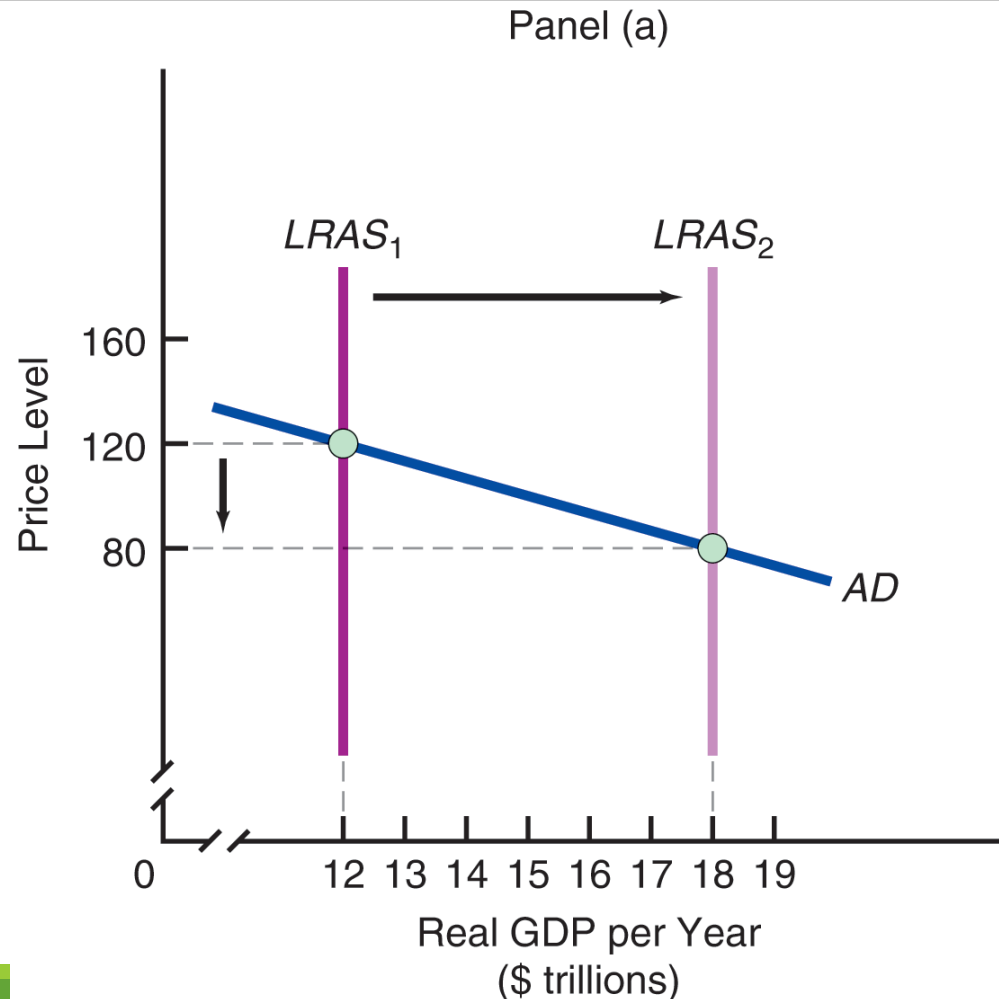
# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL

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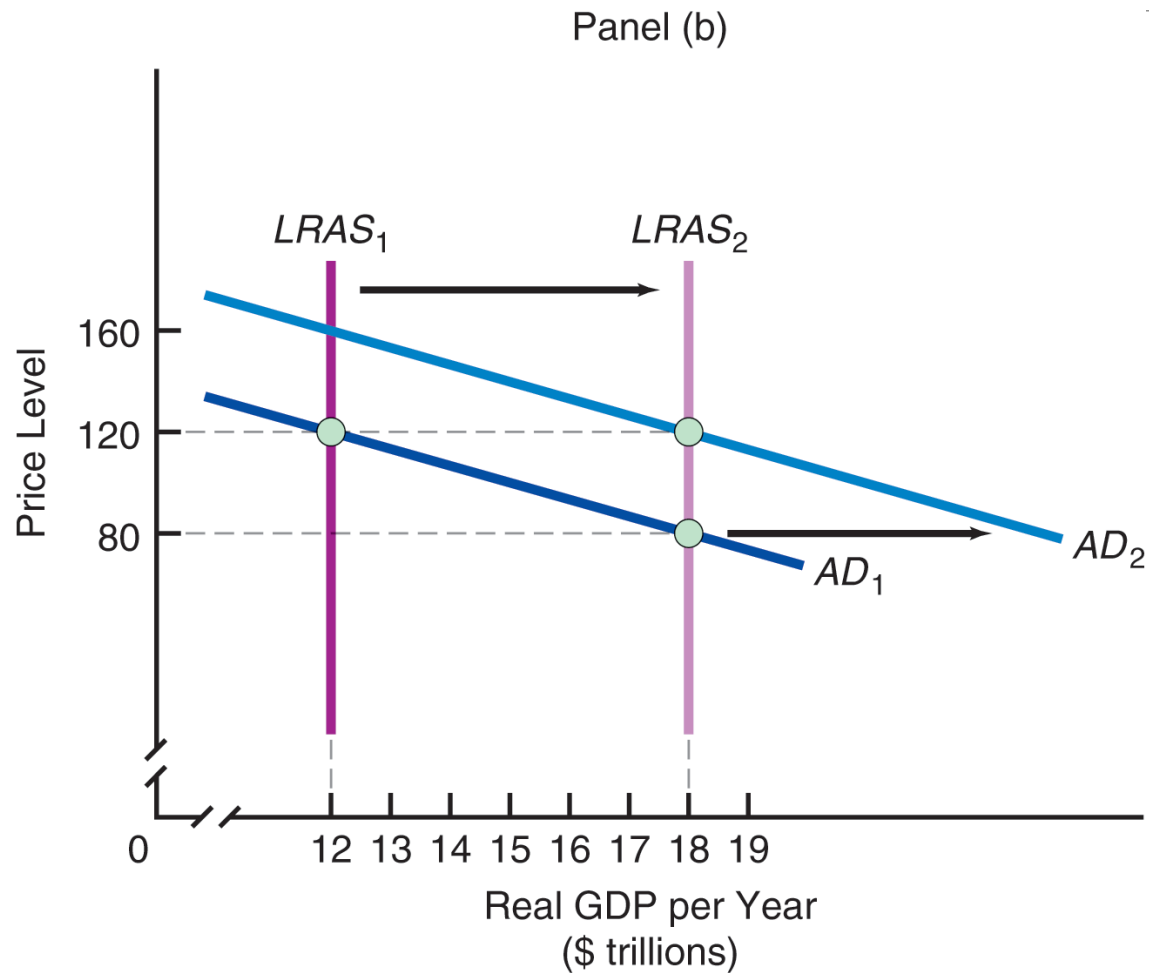
## Avoiding secular deflation

- If the *AD* curve shifts outward by the same amount as the *LRAS* curve, the price level remains constant.
  - The *AD* curve can be shifted outward by increasing the money supply.

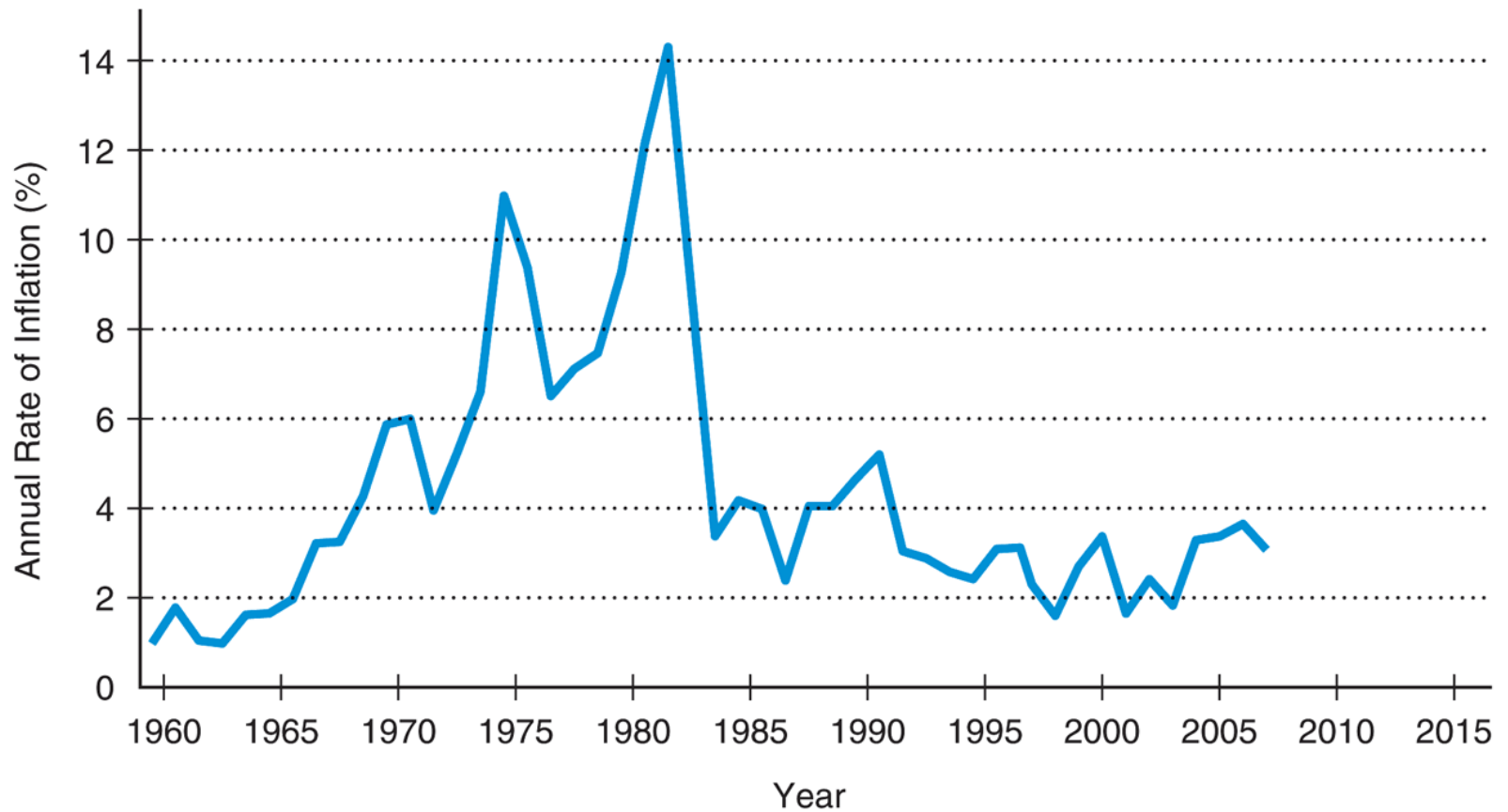
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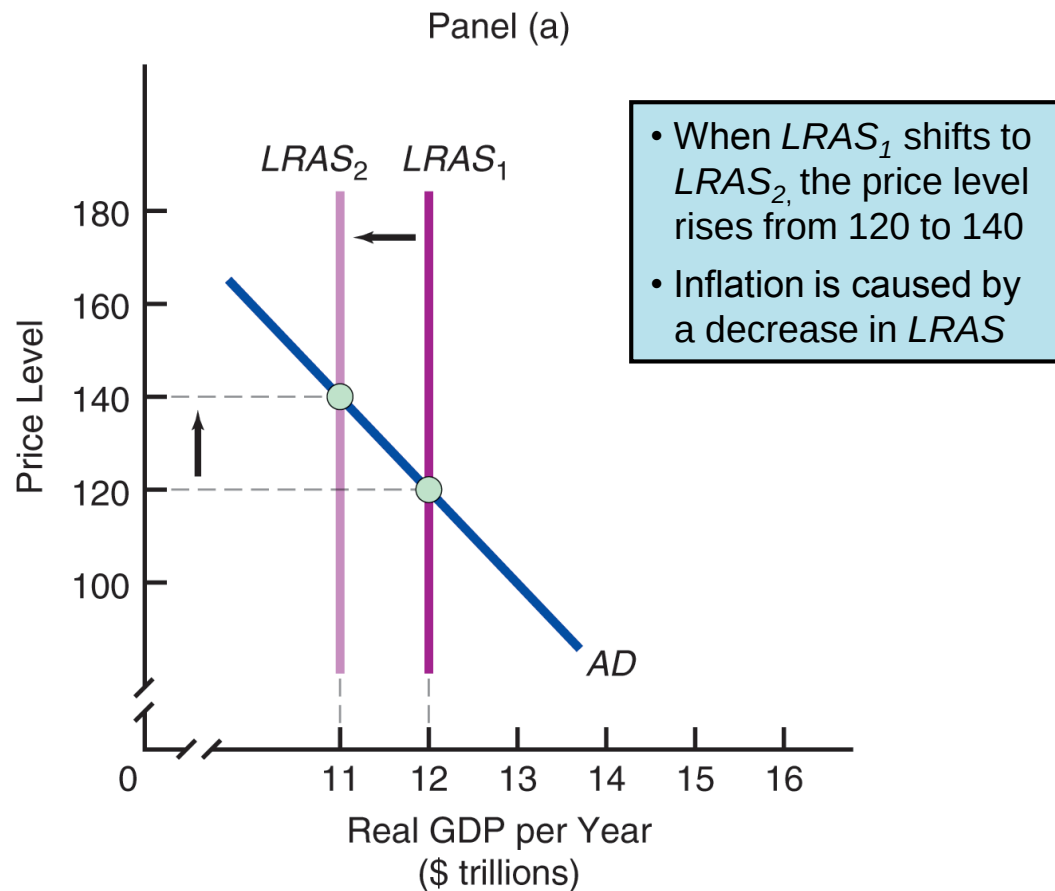
# LONG RUN EQUILIBRIUM AND THE PRICE LEVEL



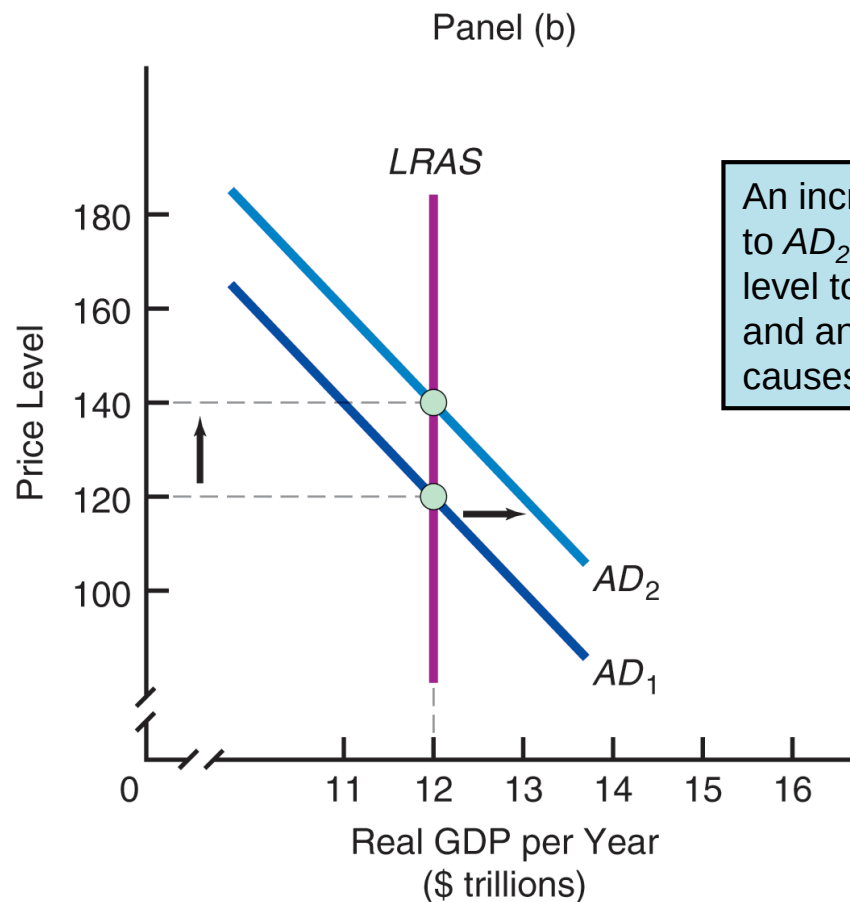
# CAUSES OF INFLATION



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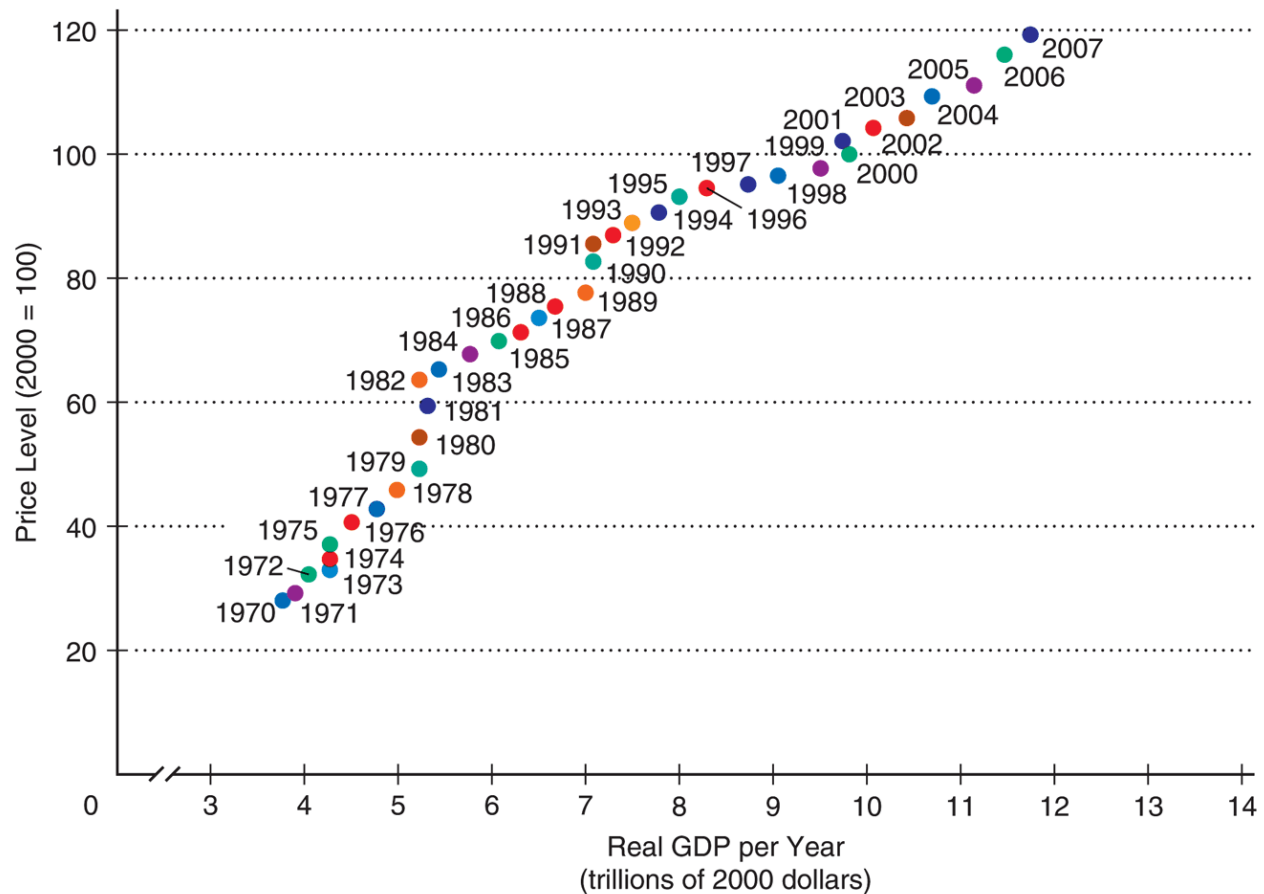


# CAUSES OF INFLATION



An increase in  $AD$  from  $AD_1$  to  $AD_2$  causes the price level to rise from 120 to 140, and an increase in  $AD$  causes inflation

# CAUSES OF INFLATION



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# END OF CHAPTER 7